Module 1: Basic Concepts in R

Lesson 2 | Objects, vectors, and data frames

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Reduced teaching slides (PDF) · PowerPoint · Module overview

This extended lesson provides the full explanation and reproducible examples behind the slides.

Hello World Program

Write the “Hello World” program using both (1) R command prompt and (2) writing a script.

R Command Prompt

Once you have R environment setup, then it’s easy to start your R command prompt by just typing the following command at your command prompt:

```
$ R
```

This will launch R interpreter and you will get a prompt > where you can start typing your program as follows:

```
# print() displays the supplied value or the contents of a saved object.
# Use <- to assign a result to a name before reusing it in later expressions.
myString <- "Hello, World!"
print (myString)
```

```
[1] "Hello, World!"
```

Here first statement defines a string variable myString, where we assign a string “Hello, World!” and then next statement print() is being used to print the value stored in variable myString.

Note The bracketed number [1] that appears before the output indicates that the line begins with the first value of the result. Some results may have multiple values that fill several lines, so this indicator is occasionally useful but can generally be ignored.

Writing R Script File

Usually, you will do your programming by writing your programs in script files and then you execute those scripts at your command prompt with the help of R interpreter called Rscript.

So let's start with writing following code in a text file called test.R:

```
# print() displays the supplied value or the contents of a saved object.
# Use <- to assign a result to a name before reusing it in later expressions.
# My first program in R Programming
myString <- "Hello, World!"
print ( myString)
```

```
[1] "Hello, World!"
```

Save the above code in a file test.R and execute it at Linux command prompt as given below.

Even if you are using Windows or other system, syntax will remain same.

```
$ Rscript test.R
```

Comments

Comments are like helping text in your R program and they are ignored by the interpreter while executing your actual program. Single comment is written using # in the beginning of the statement as follows:

```
# My first program in R Programming
```

Variables and Basic Data Types

- Generally, while doing programming in any programming language, you need to use various variables to store various information.
- Variables are reserved memory locations to store values. This means that, when you create a variable you reserve some space in memory.
- You may need to store information of various data types like character, wide character, integer, floating point, double floating point, Boolean etc.
- Based on the data type of a variable, the operating system allocates memory and decides what can be stored in the reserved memory.
- In contrast to other programming languages like C and java in R, the variables are not declared as some data type. The variables are assigned with R-Objects and the data type of the R-object becomes the data type of the variable. There are many types of R-objects.
- The frequently used ones are – **Vectors Lists - Matrices - Arrays - Factors - Data Frames**

Numeric Variables

Numeric variables are the most common type of variable and are used to store real numbers.

- **Integer**: A subtype of numeric that represents whole numbers.

```r
# Assign the expression on the right to the object named on the left.
# Later calculations use the stored value; assignment replaces any previous value with
 ↪  that name.
# Integer variable
count <- 42L # The 'L' specifies it as an integer

# Floating-point number
weight <- 72.5
```

- **Complex Numbers**: Numbers with a real and an imaginary part.

```r
# The suffix i denotes the imaginary component of a complex number in R.
# This example stores the value with real part 3 and imaginary part 4.
# Complex number
complex_num <- 3+4i
```

- **Character Variables**

Character variables store text and are enclosed in quotes.

```r
# Assign the expression on the right to the object named on the left.
# Later calculations use the stored value; assignment replaces any previous value with
 ↪  that name.
# Single character string
name <- "Alice"

# Multiple characters string
sentence <- "Hello, World!"
```

- **Logical Variables**

Logical variables can only take two values: TRUE or FALSE. They are the result of conditions or logical operations.

```r
# Logical values record TRUE or FALSE; comparisons create logical results.
# & requires both conditions, while | accepts either; prefer TRUE and FALSE over the
 ↪  reassignable T and F names.
# Logical variable
is_valid <- TRUE

# Result of a logical operation
test_result <- 5 > 3 # Returns TRUE

a <- T
b <- F

a | b
```

```
[1] TRUE
```

```r
a & b
```

```
[1] FALSE
```

R is Case Sensitive

The R language is case-sensitive, so typing the command as Print( ) or PRINT( ) will simply produce an error.

```r
# print(): displays the supplied value or object.
# Print(): has intentionally incorrect capitalization in this error example; R requires
 ↪  print().
# R is case sensitive: age, Age, and the function names print and Print are different
 ↪  names.
# The mismatched names below intentionally demonstrate errors rather than a working
 ↪  calculation.
age <- 25
print(age)
```

```
[1] 25
```

```r
print(Age)
```

```
Error:
! object 'Age' not found
```

```r
Print(age) # P is capitalized in print
```

```
Error in `Print()`:
! could not find function "Print"
```

Get Data Type of a Variable

```r
# c(): combines values into a vector, converting them to a common type when needed.
# data.frame() combines equally aligned vectors into a rectangular table.
# Columns describe variables and rows describe observations; retain the meaning and units
 ↪  of each column.
BMI <- data.frame(
gender = c("Male", "Male","Female"),
height = c(152, 171.5, 165),
weight = c(81,93, 78),
Age = c(42,38,26)
)
```

```r
# print(): displays the supplied value or object.
# str() displays structure, including column types and example values.
# class() reports the object class, while typeof() describes its underlying storage type.
print("The class of BMI is:")
```

```
[1] "The class of BMI is:"
```

```r
class(BMI)
```

```
[1] "data.frame"
```

```r
print("The type of BMI is:")
```

```
[1] "The type of BMI is:"
```

```r
typeof(BMI)
```

```
[1] "list"
```

```r
print("The structure of BMI is:")
```

```
[1] "The structure of BMI is:"
```

```r
str(BMI)
```

```
'data.frame':   3 obs. of  4 variables:
 $ gender: chr  "Male" "Male" "Female"
 $ height: num  152 172 165
 $ weight: num  81 93 78
 $ Age   : num  42 38 26
```

Logical operators in R:

<	less than
>	great than
<=	less than or equal
>=	greater than or equal
==	equal to
!=	not equal to
|	or
!	not
&	and

Complext Data Types and Data Structures

Vectors

Vectors are one-dimensional arrays that can hold numeric, character, or logical values, but all elements must be of the same type.

Patient ID	Height (CM)	Weight (KG)	Systolic (mmHg)	Diastolic (mmHg)
10200	165	71	120	80
3600	180	92	110	85
8437	172	76	118	82
5911	167	82	115	78

```r
# c(): combines values into a vector, converting them to a common type when needed.
# Logical values record TRUE or FALSE; comparisons create logical results.
# & requires both conditions, while | accepts either; prefer TRUE and FALSE over the
 ↪  reassignable T and F names.
# Numeric vector
ages <- c(25, 30, 45)

# Character vector
colors <- c("red", "green", "blue")

# Logical vector
answers <- c(TRUE, FALSE, TRUE)
```

Creating Sequences in R

Sequences are ordered sets of numbers that follow a specific pattern.

In R, sequences are commonly used for tasks such as iteration in loops, subsetting data structures, and during plotting.

R provides several functions to create sequences efficiently.

Using the Colon Operator :

The colon operator is the simplest way to create a sequence of numbers in R. It generates a sequence from a starting point to an endpoint with a step of 1.

Syntax:

```
# The colon operator creates a sequence in steps of one from start toward end.
# This is a syntax template; replace start and end with actual endpoint values.
start:end
```

Example:

Create a sequence from 1 to 10.

```
# print() displays the supplied value or the contents of a saved object.
# Use <- to assign a result to a name before reusing it in later expressions.
seq_1_to_10 <- 1:10
print(seq_1_to_10)
```

```
 [1]  1  2  3  4  5  6  7  8  9 10
```

The `seq()` Function

The `seq()` function is more versatile than the colon operator and can create sequences with specified increments (the `by` argument) or of a specified length (the `length.out` argument).

Syntax:

```
# seq() creates a sequence: from/to set endpoints, by sets spacing, and length.out sets
 ↪  the number of values.
# This is a syntax template; normally specify by or length.out, not conflicting values for
 ↪  both.
seq(from, to, by, length.out)
```

Examples:

- Create a sequence from 1 to 10 with a step of 2:

```
# print(): displays the supplied value or object.
# seq() creates evenly spaced values between the specified limits.
# by sets the step size; length.out instead requests a chosen number of evenly spaced
 ↪  values.
seq_by_2 <- seq(from = 1, to = 10, by = 2)
print(seq_by_2)
```

```
[1] 1 3 5 7 9
```

- Create a sequence from 5 to 10, consisting of 3 equally spaced numbers:

```r
# print(): displays the supplied value or object.
# seq() creates evenly spaced values between the specified limits.
# by sets the step size; length.out instead requests a chosen number of evenly spaced
 ↪  values.
seq_length_out <- seq(from = 5, to = 10, length.out = 3)
print(seq_length_out)
```

```
[1]  5.0  7.5 10.0
```

The rep() Function

The rep() function is used to replicate the values in x. It is a powerful function for repeating sequences.

Syntax:

```r
# rep() repeats values: times repeats the vector, each repeats its elements, and
 ↪  length.out sets output length.
# This is a syntax template; supply x and the repetition arguments needed for the task.
rep(x, times, each, length.out)
```

Examples:

- Repeat the sequence of numbers 1, 2, 3, three times:

```r
# print(): displays the supplied value or object.
# rep() repeats supplied values to construct a vector.
# times repeats the entire sequence; each repeats individual elements before moving on.
rep_times <- rep(x = 1:3, times = 3)
print(rep_times)
```

```
[1] 1 2 3 1 2 3 1 2 3
```

- Repeat each number in the sequence 1, 2, 3, three times:

```r
# print(): displays the supplied value or object.
# rep() repeats supplied values to construct a vector.
# times repeats the entire sequence; each repeats individual elements before moving on.
rep_each <- rep(x = 1:3, each = 3)
print(rep_each)
```

```
[1] 1 1 1 2 2 2 3 3 3
```

Combining Sequences

Sequences can be combined using the c() function to create longer and more complex sequences.

Example:

Combine two sequences into one.

```r
# c(): combines values into a vector, converting them to a common type when needed.
# print(): displays the supplied value or object.
# seq() creates evenly spaced values between the specified limits.
# by sets the step size; length.out instead requests a chosen number of evenly spaced
 ↪  values.
combined_seq <- c(seq(1, 5), seq(8, 10))
print(combined_seq)
```

```
[1]  1  2  3  4  5  8  9 10
```

Practice Vectors

- Create a vector that combines the numbers 1, 2, and 3 with the letters ‘a’, ‘b’, and ‘c’.
- Create a vector containing the numbers 1 to 10, but only include every second number.
- Create a vector that contains the sequence 1, 2, 3, followed by the sequence 4, 5, 6, each sequence repeated twice.
- Create a vector that starts with the sequence 1, 2, 3 and then repeats the number 4 five times.
- Create a vector containing the first 10 even numbers.
- Create a vector containing the first 10 odd numbers.
- Create a vector that uses the `seq()` function to generate numbers from 1 to 10, but each number should appear twice.
- Create a vector that uses the `rep()` function to repeat the sequence 1, 2, 3, but the entire sequence should appear three times.

Access Vector Elements

Accessing elements from a vector in R is straightforward and can be achieved using indexing with square brackets `[ ]`.

You can specify the index or indices of the elements you want to extract. Remember, R is 1-based indexing, meaning the first element of a vector has an index of 1.

Single Element Access

To access a single element from a vector, you provide its index within square brackets immediately following the vector name.

```r
# print(): displays the supplied value or object.
# [] selects vector values by position; R starts counting positions at 1.
# c(1,3,5) selects several positions, while 2:4 selects a consecutive range.
# Creating a numeric vector
numbers <- c(10, 20, 30, 40, 50)

# Access the third element
third_number <- numbers[3]
print(third_number)  # Outputs 30
```

```
[1] 30
```

Multiple Elements Access

To access multiple elements, you can pass a vector of indices inside the square brackets.

```r
# print(): displays the supplied value or object.
# [] selects vector values by position; R starts counting positions at 1.
# c(1,3,5) selects several positions, while 2:4 selects a consecutive range.
# Access the first, third, and fifth elements
selected_numbers <- numbers[c(1, 3, 5)]
print(selected_numbers)  # Outputs 10, 30, 50
```

```
[1] 10 30 50
```

Sequential Elements Access

R allows you to use the colon `:` operator to create a sequence, which can be used for accessing a range of elements.

```r
# print(): displays the supplied value or object.
# [] selects vector values by position; R starts counting positions at 1.
# c(1,3,5) selects several positions, while 2:4 selects a consecutive range.
# Access the second to the fourth elements
range_numbers <- numbers[2:4]
print(range_numbers)  # Outputs 20, 30, 40
```

```
[1] 20 30 40
```

Conditional Access

You can also access elements based on conditions. This method is useful for filtering elements.

```r
# print(): displays the supplied value or object.
# A logical condition inside [] selects the entries for which it is TRUE.
# The result contains matching values in their original order, rather than their integer
 ↪  positions.
# Access elements greater than 25
filtered_numbers <- numbers[numbers > 25]
print(filtered_numbers)  # Outputs 30, 40, 50
```

```
[1] 30 40 50
```

Find the Index of Matching Elements

```r
# which() returns the positions where a logical condition is TRUE.
# Use these integer positions to select matching entries without changing the original
 ↪  vector.
which(numbers > 25)
```

```
[1] 3 4 5
```

Excluding Elements

Adding a minus sign `-` before an index or a vector of indices tells R to exclude those elements.

```r
# print(): displays the supplied value or object.
# Negative indices inside [] exclude the specified positions from a vector.
# Parentheses keep a multi-position exclusion together, as in -(2:4).
# Exclude the second element
exclude_second <- numbers[-2]
print(exclude_second)  # Outputs 10, 30, 40, 50
```

```
[1] 10 30 40 50
```

```r
# Exclude second to fourth elements
exclude_range <- numbers[-(2:4)]
print(exclude_range)  # Outputs 10, 50
```

```
[1] 10 50
```

Named Vectors

For named vectors, you can access elements using their names in a similar manner.

```
# print(): displays the supplied value or object.
# c() combines compatible values into a vector in the supplied order.
# Assignments save the vector under a name; printing that name displays its current
 ↪  values.
# Creating a named vector
fruits <- c(apple = 10, banana = 5, cherry = 8)

# Access the element named 'banana'
banana_count <- fruits["banana"]
print(banana_count)  # Outputs 5
```

```
banana
     5
```

```
# Access multiple elements by name
some_fruits <- fruits[c("apple", "cherry")]
print(some_fruits)  # Outputs apple 10, cherry 8
```

```
 apple cherry
    10      8
```

Basic Operations on Vectors

- Once you have a vector (or a list of numbers) in memory most basic operations are available.
- Most of the basic operations will act on a whole vector and can be used to quickly perform a large number of calculations with a single command.
- There is one thing to note, if you perform an operation on more than one vector it is often necessary that the vectors all contain the same number of entries.
- Here we first define a vector which we will call “a” and will look at how to add and subtract constant numbers from all of the numbers in the vector.
- First, the vector will contain the numbers 1, 2, 3, and 4.
- We then see how to add 5 to each of the numbers, subtract 10 from each of the numbers, multiply each number by 4, and divide each number by 5.

```
# c() combines compatible values into a vector in the supplied order.
# Assignments save the vector under a name; printing that name displays its current
 ↪  values.
a <- c(1,2,3,4)
a
```

```
[1] 1 2 3 4
```

```
a <- a + 5
a
```

```
[1] 6 7 8 9
```

```
a <- a - 10
a
```

```
[1] -4 -3 -2 -1
```

```
a <- a / 5
a
```

```
[1] -0.8 -0.6 -0.4 -0.2
```

We can save the results in another vector called b:

```
# Assign the expression on the right to the object named on the left.
# Later calculations use the stored value; assignment replaces any previous value with
 ↪  that name.
b <- a
b
```

```
[1] -0.8 -0.6 -0.4 -0.2
```

If you want to raise to a power, take the square root, find e raised to each number, the logarithm, etc., then the usual commands can be used:

```
# c(): combines values into a vector, converting them to a common type when needed.
# exp(): computes e raised to each supplied value.
# log(): computes natural logarithms by default; base selects another logarithm base.
# sqrt() returns the square root of each input value.
# Arithmetic on vectors is element by element; check domains before taking square roots or
 ↪  logarithms.
a <- c(1,2,3,4)
a ^ 2
```

```
[1]  1  4  9 16
```

```
sqrt(a)
```

```
[1] 1.000000 1.414214 1.732051 2.000000
```

```
exp(a)
```

```
[1]  2.718282  7.389056 20.085537 54.598150
```

```
log(a)
```

```
[1] 0.0000000 0.6931472 1.0986123 1.3862944
```

```
exp(log(a))
```

```
[1] 1 2 3 4
```

By combining operations and using parentheses you can make more complicated expressions:

```
# exp(): computes e raised to each supplied value.
# sqrt() returns the square root of each input value.
# Arithmetic on vectors is element by element; check domains before taking square roots or
 ↪  logarithms.
c <- (a + sqrt(a))/(exp(2)+1)
c
```

```
[1] 0.2384058 0.4069842 0.5640743 0.7152175
```

Note that you can do the same operations with vector arguments.

For example to add the elements in vector a to the elements in vector b use the following command:

```
# Apply the arithmetic operation element by element to the supplied vectors.
# Unequal vector lengths invoke recycling, so check lengths before interpreting the
 ↪  result.
a + b
```

```
[1] 0.2 1.4 2.6 3.8
```

The operation is performed on an element by element basis.

Note this is true for almost all of the basic functions.

So you can bring together all kinds of complicated expressions:

```
# sqrt() returns the square root of each input value.
# Arithmetic on vectors is element by element; check domains before taking square roots or
 ↪  logarithms.
a*b
```

```
[1] -0.8 -1.2 -1.2 -0.8
```

```
a/b
```

```
[1]  -1.250000  -3.333333  -7.500000 -20.000000
```

```
(a+3)/(sqrt(1-b)*2-1)
```

```
[1] 2.376311 3.268354 4.390998 5.877956
```

You need to be careful of one thing. When you do operations on vectors they are performed on an element by element basis. One ramification of this is that all of the vectors in an expression must be the same length. If the lengths of the vectors differ then you may get an error message, or worse, a warning message and unpredictable results:

```
# c() combines compatible values into a vector in the supplied order.
# Assignments save the vector under a name; printing that name displays its current
 ↪  values.
a <- c(1,2,3)
b <- c(10,11,12,13)
a+b
```

Basic Numerical Descriptors

The following commands can be used to get the mean, median, quantiles, minimum, maximum, variance, and standard deviation of a set of numbers:

```
# quantile(): computes sample percentiles; probs gives probabilities and type selects the
 ↪  interpolation rule.
# min(): returns the smallest supplied value.
# max(): returns the largest supplied value.
# mean() and median() describe center; var() and sd() summarize numerical spread.
# var() uses the sample denominator n-1; sd() expresses that spread in the original units.
a <- 1:10
mean(a)
```

```
[1] 5.5
```

```
median(a)
```

```
[1] 5.5
```

```
quantile(a)
```

```
   0%   25%   50%   75%  100%
 1.00  3.25  5.50  7.75 10.00
```

```r
var(a)
```

```
[1] 9.166667
```

```r
sd(a)
```

```
[1] 3.02765
```

```r
min(a)
```

```
[1] 1
```

```r
max(a)
```

```
[1] 10
```

Finally, the summary command will print out the min, max, mean, median, and quantiles:

```r
# summary() reports an overview appropriate to the object or its columns.
# For numerical vectors it shows the range, quartiles, median, and mean; inspect the
 ↪  object type too.
summary(a)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
   1.00    3.25    5.50    5.50    7.75   10.00
```

The summary command is especially nice because if you give it a data frame it will print out the summary for every vector in the data frame:

```r
# summary() reports an overview appropriate to the object or its columns.
# For numerical vectors it shows the range, quartiles, median, and mean; inspect the
 ↪  object type too.
summary(mtcars)
```

```
      mpg             cyl             disp             hp
 Min.   :10.40   Min.   :4.000   Min.   : 71.1   Min.   : 52.0
 1st Qu.:15.43   1st Qu.:4.000   1st Qu.:120.8   1st Qu.: 96.5
 Median :19.20   Median :6.000   Median :196.3   Median :123.0
 Mean   :20.09   Mean   :6.188   Mean   :230.7   Mean   :146.7
 3rd Qu.:22.80   3rd Qu.:8.000   3rd Qu.:326.0   3rd Qu.:180.0
 Max.   :33.90   Max.   :8.000   Max.   :472.0   Max.   :335.0
      drat             wt             qsec             vs
 Min.   :2.760   Min.   :1.513   Min.   :14.50   Min.   :0.0000
 1st Qu.:3.080   1st Qu.:2.581   1st Qu.:16.89   1st Qu.:0.0000
 Median :3.695   Median :3.325   Median :17.71   Median :0.0000
 Mean   :3.597   Mean   :3.217   Mean   :17.85   Mean   :0.4375
 3rd Qu.:3.920   3rd Qu.:3.610   3rd Qu.:18.90   3rd Qu.:1.0000
 Max.   :4.930   Max.   :5.424   Max.   :22.90   Max.   :1.0000
       am              gear            carb
 Min.   :0.0000   Min.   :3.000   Min.   :1.000
 1st Qu.:0.0000   1st Qu.:3.000   1st Qu.:2.000
 Median :0.0000   Median :4.000   Median :2.000
 Mean   :0.4062   Mean   :3.688   Mean   :2.812
 3rd Qu.:1.0000   3rd Qu.:4.000   3rd Qu.:4.000
 Max.   :1.0000   Max.   :5.000   Max.   :8.000
```

Sorting Vectors

Here we look at some commonly used commands that perform operations on lists.

The commands include the sort, min, max, and sum commands.

First, the sort command can sort the given vector in either ascending or descending order:

```r
# c(): combines values into a vector, converting them to a common type when needed.
# sort() rearranges values into increasing order by default.
# decreasing=TRUE reverses the order; this returns values rather than their original
 ↪  positions.
a = c(2,4,6,3,1,5)
b = sort(a)
c = sort(a,decreasing = TRUE)
```

Speed of Vectorized Operations

Using vectorization to multiply numbers vs using for loop

```r
# rnorm(): generates artificial normal draws; n is the draw count and sd is standard
 ↪  deviation.
# rep(): repeats values; times repeats a vector and each repeats individual elements.
# A for loop repeats its body once for each value in the supplied sequence.
# The loop variable identifies the current value or position; vectorized arithmetic can
 ↪  often replace the loop.
N <- 100
a <- rnorm(N)
b <- rnorm(N)

# Vectorized approach
c <- a * b

# De-vectorized approach
d <- rep(NA, N)
for(i in 1:N){
  d[i] <- a[i] * b[i]
}
```

```r
# rnorm(): generates artificial normal draws; n is the draw count and sd is standard
 ↪  deviation.
# system.time(): measures the computing and elapsed time needed to evaluate an expression.
# rep(): repeats values; times repeats a vector and each repeats individual elements.
# print(): displays the supplied value or object.
# A for loop repeats its body once for each value in the supplied sequence.
# The loop variable identifies the current value or position; vectorized arithmetic can
 ↪  often replace the loop.
# Use 100,000 values to keep memory use modest on student computers.
# Timings vary by computer and run; compare approaches rather than expecting fixed
 ↪  results.
N <- 100000
a <- rnorm(N)
b <- rnorm(N)

# Measure the time for the vectorized approach
vectorized_time <- system.time({
  c <- a * b
})

# Measure the time for the de-vectorized approach
de_vectorized_time <- system.time({
  d <- rep(NA, N)
  for(i in 1:N){
    d[i] <- a[i] * b[i]
  }
})

# Print the timings
print(vectorized_time)
```

```
   user  system elapsed
  0.001   0.000   0.000
```

```
print(de_vectorized_time)
```

```
   user  system elapsed
  0.003   0.001   0.004
```

Other Data Structures (Matrix, Array, List and Data frames)

Matrices

Matrices are two-dimensional arrays that contain elements of the same type. They have rows and columns.

```
# matrix() arranges values into rows and columns of one underlying data type.
# R fills columns first by default; nrow and ncol specify the requested dimensions.
# Matrix with 3 rows and 2 columns
matrix_data <- matrix(1:6, nrow = 3, ncol = 2)
```

Note: Check the NBA Example Quarto File

Arrays

Arrays are similar to matrices but can have more than two dimensions.

```
# c(): combines values into a vector, converting them to a common type when needed.
# array() arranges values into a multidimensional object of one underlying type.
# dim specifies the size of each dimension, in the order used when indexing the array.
# 3-dimensional array
array_data <- array(1:24, dim = c(3, 4, 2))
```

Lists

Lists can hold elements of different types, including numbers, strings, vectors, and even other lists.

```
# c(): combines values into a vector, converting them to a common type when needed.
# print(): displays the supplied value or object.
# list() keeps several objects together even when their types or lengths differ.
# Single brackets return a sublist; double brackets extract an element, and $ can select a
 ↪  named element.
# List containing different types
list_data <- list(name = "Alice", age = 25, scores = c(90, 80, 85))

print(list_data[1])
```

```
$name
[1] "Alice"
```

Data Frames

Data frames are used to store tabular data. They are similar to matrices but can contain different types of variables.

```
# c(): combines values into a vector, converting them to a common type when needed.
# data.frame() combines equally aligned vectors into a rectangular table.
# Columns describe variables and rows describe observations; retain the meaning and units
 ↪  of each column.
# Data frame
df <- data.frame(
  Name = c("Alice", "Bob"),
  Age = c(25, 30),
  Weight = c(55.5, 85.7)
)
```

Access Data Frame Elements

R built-in datasets

R provides several datasets that can be used for practice.

Use the data() command to examine these datasets.

```
#data()

# we will use mtcars dataset for demonstration
#?mtcars
#View(trees)
```

Using $ to Access Columns

The **$** operator is used to access a column directly by its name, returning the column as a vector.

```
# print(): displays the supplied value or object.
# mpg in mtcars measures miles per gallon, not miles per hour.
# $ extracts a column vector; single-column ["mpg"] indexing retains a data frame.
miles_per_hour <- mtcars$mpg
print(miles_per_hour)
```

```
 [1] 21.0 21.0 22.8 21.4 18.7 18.1 14.3 24.4 22.8 19.2 17.8 16.4 17.3 15.2 10.4
[16] 10.4 14.7 32.4 30.4 33.9 21.5 15.5 15.2 13.3 19.2 27.3 26.0 30.4 15.8 19.7
[31] 15.0 21.4
```

Using [] for Row and Column Indexing

The **[ ]** operator allows for more flexible sub-setting, including selecting rows, columns, or both, using numeric indexes or column names.

- **Select a single column by name**:

```
# print(): displays the supplied value or object.
# mpg in mtcars measures miles per gallon, not miles per hour.
# $ extracts a column vector; single-column ["mpg"] indexing retains a data frame.
miles_per_hour <- mtcars["mpg"]
print(miles_per_hour)
```

```
                   mpg
Mazda RX4         21.0
Mazda RX4 Wag     21.0
Datsun 710        22.8
Hornet 4 Drive    21.4
Hornet Sportabout 18.7
Valiant           18.1
Duster 360        14.3
Merc 240D         24.4
Merc 230          22.8
Merc 280          19.2
Merc 280C         17.8
Merc 450SE        16.4
Merc 450SL        17.3
Merc 450SLC       15.2
Cadillac Fleetwood  10.4
Lincoln Continental 10.4
Chrysler Imperial   14.7
Fiat 128          32.4
Honda Civic       30.4
Toyota Corolla    33.9
Toyota Corona     21.5
Dodge Challenger  15.5
AMC Javelin       15.2
Camaro Z28        13.3
Pontiac Firebird  19.2
Fiat X1-9         27.3
Porsche 914-2     26.0
Lotus Europa      30.4
Ford Pantera L    15.8
Ferrari Dino      19.7
Maserati Bora     15.0
Volvo 142E        21.4
```

Select multiple columns by name

```r
# c(): combines values into a vector, converting them to a common type when needed.
# print(): displays the supplied value or object.
# A data frame uses [row, column] indexing to select records and variables.
# A blank row or column position keeps all entries in that dimension.
subset_df <- mtcars[c("mpg", "cyl")]
print(subset_df)
```

```
                   mpg cyl
Mazda RX4         21.0   6
Mazda RX4 Wag     21.0   6
Datsun 710        22.8   4
Hornet 4 Drive    21.4   6
Hornet Sportabout 18.7   8
Valiant           18.1   6
Duster 360        14.3   8
Merc 240D         24.4   4
Merc 230          22.8   4
Merc 280          19.2   6
Merc 280C         17.8   6
Merc 450SE        16.4   8
Merc 450SL        17.3   8
Merc 450SLC       15.2   8
Cadillac Fleetwood  10.4   8
Lincoln Continental 10.4   8
Chrysler Imperial   14.7   8
Fiat 128          32.4   4
Honda Civic       30.4   4
Toyota Corolla    33.9   4
Toyota Corona     21.5   4
Dodge Challenger  15.5   8
AMC Javelin       15.2   8
```

```
Camaro Z28          13.3   8
Pontiac Firebird    19.2   8
Fiat X1-9           27.3   4
Porsche 914-2       26.0   4
Lotus Europa        30.4   4
Ford Pantera L      15.8   8
Ferrari Dino        19.7   6
Maserati Bora       15.0   8
Volvo 142E          21.4   4
```

```r
# print(): displays the supplied value or object.
# head(): shows the first n entries or rows, six by default.
# c(): combines values into a vector, converting them to a common type when needed.
# length(): counts elements in a vector or list.
# nrow(): returns the number of rows in a table or matrix.
# ncol(): returns the number of columns in a table or matrix.
# mean(): computes an average; logical TRUE/FALSE values contribute 1/0, giving a
 ↪  proportion.
# mpg in mtcars measures miles per gallon, not miles per hour.
# $ extracts a column vector; single-column ["mpg"] indexing retains a data frame.
# In R, rownames is a function that retrieves or sets the row names of a matrix or # data
 ↪  frame.
# rownames are essentially a character vector of unique identifiers that label the rows,
 ↪  similar to how column names label the columns.

# Get all rownames
car_names <- rownames(mtcars)
print(car_names)
```

```
 [1] "Mazda RX4"           "Mazda RX4 Wag"       "Datsun 710"
 [4] "Hornet 4 Drive"      "Hornet Sportabout"   "Valiant"
 [7] "Duster 360"          "Merc 240D"           "Merc 230"
[10] "Merc 280"            "Merc 280C"           "Merc 450SE"
[13] "Merc 450SL"          "Merc 450SLC"         "Cadillac Fleetwood"
[16] "Lincoln Continental" "Chrysler Imperial"   "Fiat 128"
[19] "Honda Civic"         "Toyota Corolla"      "Toyota Corona"
[22] "Dodge Challenger"    "AMC Javelin"         "Camaro Z28"
[25] "Pontiac Firebird"    "Fiat X1-9"           "Porsche 914-2"
[28] "Lotus Europa"        "Ford Pantera L"      "Ferrari Dino"
[31] "Maserati Bora"       "Volvo 142E"
```

```r
# Get first rowname only
first_car_name <- rownames(mtcars)[1]
print(first_car_name)
```

```
[1] "Mazda RX4"
```

```r
# Find record by rowname
datsun_710 <- mtcars["Datsun 710", ]
print(datsun_710)
```

```
           mpg cyl disp hp drat   wt  qsec vs am gear carb
Datsun 710 22.8   4  108 93 3.85 2.32 18.61  1  1    4    1
```

```r
mtcars$car_brand <- rownames(mtcars)
# rownames(mtcars) <- NULL
head(mtcars)
```

```
                  mpg cyl disp  hp drat    wt  qsec vs am gear carb
Mazda RX4         21.0   6  160 110 3.90 2.620 16.46  0  1    4    4
Mazda RX4 Wag     21.0   6  160 110 3.90 2.875 17.02  0  1    4    4
Datsun 710        22.8   4  108  93 3.85 2.320 18.61  1  1    4    1
Hornet 4 Drive    21.4   6  258 110 3.08 3.215 19.44  1  0    3    1
Hornet Sportabout 18.7   8  360 175 3.15 3.440 17.02  0  0    3    2
Valiant           18.1   6  225 105 2.76 3.460 20.22  1  0    3    1
                         car_brand
```

```
Mazda RX4                 Mazda RX4
Mazda RX4 Wag         Mazda RX4 Wag
Datsun 710               Datsun 710
Hornet 4 Drive       Hornet 4 Drive
Hornet Sportabout Hornet Sportabout
Valiant                     Valiant
```

```
#View(mtcars)

# select rows 5 to 7
mtcars
```

```
                   mpg cyl  disp  hp drat    wt  qsec vs am gear carb
Mazda RX4         21.0   6 160.0 110 3.90 2.620 16.46  0  1    4    4
Mazda RX4 Wag     21.0   6 160.0 110 3.90 2.875 17.02  0  1    4    4
Datsun 710        22.8   4 108.0  93 3.85 2.320 18.61  1  1    4    1
Hornet 4 Drive    21.4   6 258.0 110 3.08 3.215 19.44  1  0    3    1
Hornet Sportabout 18.7   8 360.0 175 3.15 3.440 17.02  0  0    3    2
Valiant           18.1   6 225.0 105 2.76 3.460 20.22  1  0    3    1
Duster 360        14.3   8 360.0 245 3.21 3.570 15.84  0  0    3    4
Merc 240D         24.4   4 146.7  62 3.69 3.190 20.00  1  0    4    2
Merc 230          22.8   4 140.8  95 3.92 3.150 22.90  1  0    4    2
Merc 280          19.2   6 167.6 123 3.92 3.440 18.30  1  0    4    4
Merc 280C         17.8   6 167.6 123 3.92 3.440 18.90  1  0    4    4
Merc 450SE        16.4   8 275.8 180 3.07 4.070 17.40  0  0    3    3
Merc 450SL        17.3   8 275.8 180 3.07 3.730 17.60  0  0    3    3
Merc 450SLC       15.2   8 275.8 180 3.07 3.780 18.00  0  0    3    3
Cadillac Fleetwood  10.4   8 472.0 205 2.93 5.250 17.98  0  0    3    4
Lincoln Continental 10.4   8 460.0 215 3.00 5.424 17.82  0  0    3    4
Chrysler Imperial   14.7   8 440.0 230 3.23 5.345 17.42  0  0    3    4
Fiat 128          32.4   4  78.7  66 4.08 2.200 19.47  1  1    4    1
Honda Civic       30.4   4  75.7  52 4.93 1.615 18.52  1  1    4    2
Toyota Corolla    33.9   4  71.1  65 4.22 1.835 19.90  1  1    4    1
Toyota Corona     21.5   4 120.1  97 3.70 2.465 20.01  1  0    3    1
Dodge Challenger  15.5   8 318.0 150 2.76 3.520 16.87  0  0    3    2
AMC Javelin       15.2   8 304.0 150 3.15 3.435 17.30  0  0    3    2
Camaro Z28        13.3   8 350.0 245 3.73 3.840 15.41  0  0    3    4
Pontiac Firebird  19.2   8 400.0 175 3.08 3.845 17.05  0  0    3    2
Fiat X1-9         27.3   4  79.0  66 4.08 1.935 18.90  1  1    4    1
Porsche 914-2     26.0   4 120.3  91 4.43 2.140 16.70  0  1    5    2
Lotus Europa      30.4   4  95.1 113 3.77 1.513 16.90  1  1    5    2
Ford Pantera L    15.8   8 351.0 264 4.22 3.170 14.50  0  1    5    4
Ferrari Dino      19.7   6 145.0 175 3.62 2.770 15.50  0  1    5    6
Maserati Bora     15.0   8 301.0 335 3.54 3.570 14.60  0  1    5    8
Volvo 142E        21.4   4 121.0 109 4.11 2.780 18.60  1  1    4    2
                            car_brand
Mazda RX4                   Mazda RX4
Mazda RX4 Wag             Mazda RX4 Wag
Datsun 710                  Datsun 710
Hornet 4 Drive           Hornet 4 Drive
Hornet Sportabout     Hornet Sportabout
Valiant                       Valiant
Duster 360                  Duster 360
Merc 240D                   Merc 240D
Merc 230                     Merc 230
Merc 280                     Merc 280
Merc 280C                   Merc 280C
Merc 450SE                  Merc 450SE
Merc 450SL                  Merc 450SL
Merc 450SLC                Merc 450SLC
Cadillac Fleetwood   Cadillac Fleetwood
Lincoln Continental Lincoln Continental
Chrysler Imperial     Chrysler Imperial
Fiat 128                       Fiat 128
Honda Civic                 Honda Civic
Toyota Corolla           Toyota Corolla
Toyota Corona             Toyota Corona
```

```
Dodge Challenger       Dodge Challenger
AMC Javelin                 AMC Javelin
Camaro Z28                   Camaro Z28
Pontiac Firebird       Pontiac Firebird
Fiat X1-9                     Fiat X1-9
Porsche 914-2              Porsche 914-2
Lotus Europa                Lotus Europa
Ford Pantera L            Ford Pantera L
Ferrari Dino                Ferrari Dino
Maserati Bora              Maserati Bora
Volvo 142E                    Volvo 142E
```

```
mtcars[5:7,]
```

```
                   mpg cyl disp  hp drat   wt  qsec vs am gear carb
Hornet Sportabout 18.7   8  360 175 3.15 3.44 17.02  0  0    3    2
Valiant           18.1   6  225 105 2.76 3.46 20.22  1  0    3    1
Duster 360        14.3   8  360 245 3.21 3.57 15.84  0  0    3    4
                          car_brand
Hornet Sportabout Hornet Sportabout
Valiant                     Valiant
Duster 360                Duster 360
```

```
# select columns 2 to 9
mtcars[2:9]
```

```
                  cyl  disp  hp drat    wt  qsec vs am
Mazda RX4           6 160.0 110 3.90 2.620 16.46  0  1
Mazda RX4 Wag       6 160.0 110 3.90 2.875 17.02  0  1
Datsun 710          4 108.0  93 3.85 2.320 18.61  1  1
Hornet 4 Drive      6 258.0 110 3.08 3.215 19.44  1  0
Hornet Sportabout   8 360.0 175 3.15 3.440 17.02  0  0
Valiant             6 225.0 105 2.76 3.460 20.22  1  0
Duster 360          8 360.0 245 3.21 3.570 15.84  0  0
Merc 240D           4 146.7  62 3.69 3.190 20.00  1  0
Merc 230            4 140.8  95 3.92 3.150 22.90  1  0
Merc 280            6 167.6 123 3.92 3.440 18.30  1  0
Merc 280C           6 167.6 123 3.92 3.440 18.90  1  0
Merc 450SE          8 275.8 180 3.07 4.070 17.40  0  0
Merc 450SL          8 275.8 180 3.07 3.730 17.60  0  0
Merc 450SLC         8 275.8 180 3.07 3.780 18.00  0  0
Cadillac Fleetwood  8 472.0 205 2.93 5.250 17.98  0  0
Lincoln Continental 8 460.0 215 3.00 5.424 17.82  0  0
Chrysler Imperial   8 440.0 230 3.23 5.345 17.42  0  0
Fiat 128            4  78.7  66 4.08 2.200 19.47  1  1
Honda Civic         4  75.7  52 4.93 1.615 18.52  1  1
Toyota Corolla      4  71.1  65 4.22 1.835 19.90  1  1
Toyota Corona       4 120.1  97 3.70 2.465 20.01  1  0
Dodge Challenger    8 318.0 150 2.76 3.520 16.87  0  0
AMC Javelin         8 304.0 150 3.15 3.435 17.30  0  0
Camaro Z28          8 350.0 245 3.73 3.840 15.41  0  0
Pontiac Firebird    8 400.0 175 3.08 3.845 17.05  0  0
Fiat X1-9           4  79.0  66 4.08 1.935 18.90  1  1
Porsche 914-2       4 120.3  91 4.43 2.140 16.70  0  1
Lotus Europa        4  95.1 113 3.77 1.513 16.90  1  1
Ford Pantera L      8 351.0 264 4.22 3.170 14.50  0  1
Ferrari Dino        6 145.0 175 3.62 2.770 15.50  0  1
Maserati Bora       8 301.0 335 3.54 3.570 14.60  0  1
Volvo 142E          4 121.0 109 4.11 2.780 18.60  1  1
```

```
mtcars[,2:9]
```

```
                  cyl  disp  hp drat    wt  qsec vs am
Mazda RX4           6 160.0 110 3.90 2.620 16.46  0  1
Mazda RX4 Wag       6 160.0 110 3.90 2.875 17.02  0  1
Datsun 710          4 108.0  93 3.85 2.320 18.61  1  1
```

```
Hornet 4 Drive        6 258.0 110 3.08 3.215 19.44  1  0
Hornet Sportabout     8 360.0 175 3.15 3.440 17.02  0  0
Valiant               6 225.0 105 2.76 3.460 20.22  1  0
Duster 360            8 360.0 245 3.21 3.570 15.84  0  0
Merc 240D             4 146.7  62 3.69 3.190 20.00  1  0
Merc 230              4 140.8  95 3.92 3.150 22.90  1  0
Merc 280              6 167.6 123 3.92 3.440 18.30  1  0
Merc 280C             6 167.6 123 3.92 3.440 18.90  1  0
Merc 450SE            8 275.8 180 3.07 4.070 17.40  0  0
Merc 450SL            8 275.8 180 3.07 3.730 17.60  0  0
Merc 450SLC           8 275.8 180 3.07 3.780 18.00  0  0
Cadillac Fleetwood    8 472.0 205 2.93 5.250 17.98  0  0
Lincoln Continental   8 460.0 215 3.00 5.424 17.82  0  0
Chrysler Imperial     8 440.0 230 3.23 5.345 17.42  0  0
Fiat 128              4  78.7  66 4.08 2.200 19.47  1  1
Honda Civic           4  75.7  52 4.93 1.615 18.52  1  1
Toyota Corolla        4  71.1  65 4.22 1.835 19.90  1  1
Toyota Corona         4 120.1  97 3.70 2.465 20.01  1  0
Dodge Challenger      8 318.0 150 2.76 3.520 16.87  0  0
AMC Javelin           8 304.0 150 3.15 3.435 17.30  0  0
Camaro Z28            8 350.0 245 3.73 3.840 15.41  0  0
Pontiac Firebird      8 400.0 175 3.08 3.845 17.05  0  0
Fiat X1-9             4  79.0  66 4.08 1.935 18.90  1  1
Porsche 914-2         4 120.3  91 4.43 2.140 16.70  0  1
Lotus Europa          4  95.1 113 3.77 1.513 16.90  1  1
Ford Pantera L        8 351.0 264 4.22 3.170 14.50  0  1
Ferrari Dino          6 145.0 175 3.62 2.770 15.50  0  1
Maserati Bora         8 301.0 335 3.54 3.570 14.60  0  1
Volvo 142E            4 121.0 109 4.11 2.780 18.60  1  1
```

```
mtcars[2,9]
```

```
[1] 1
```

```
mtcars[2:9,]
```

```
                  mpg cyl  disp  hp drat    wt  qsec vs am gear carb
Mazda RX4 Wag     21.0   6 160.0 110 3.90 2.875 17.02  0  1    4    4
Datsun 710        22.8   4 108.0  93 3.85 2.320 18.61  1  1    4    1
Hornet 4 Drive    21.4   6 258.0 110 3.08 3.215 19.44  1  0    3    1
Hornet Sportabout 18.7   8 360.0 175 3.15 3.440 17.02  0  0    3    2
Valiant           18.1   6 225.0 105 2.76 3.460 20.22  1  0    3    1
Duster 360        14.3   8 360.0 245 3.21 3.570 15.84  0  0    3    4
Merc 240D         24.4   4 146.7  62 3.69 3.190 20.00  1  0    4    2
Merc 230          22.8   4 140.8  95 3.92 3.150 22.90  1  0    4    2
                         car_brand
Mazda RX4 Wag         Mazda RX4 Wag
Datsun 710               Datsun 710
Hornet 4 Drive       Hornet 4 Drive
Hornet Sportabout Hornet Sportabout
Valiant                     Valiant
Duster 360               Duster 360
Merc 240D                 Merc 240D
Merc 230                   Merc 230
```

```
# select rows 20 to 30 and columns 8 to 10
mtcars[c(20:30),c(8:10)]
```

```
                vs am gear
Toyota Corolla   1  1    4
Toyota Corona    1  0    3
Dodge Challenger 0  0    3
AMC Javelin      0  0    3
Camaro Z28       0  0    3
Pontiac Firebird 0  0    3
Fiat X1-9        1  1    4
```

```
Porsche 914-2     0  1    5
Lotus Europa      1  1    5
Ford Pantera L    0  1    5
Ferrari Dino      0  1    5
```

```
mtcars[20:30,8:10]
```

```
                 vs am gear
Toyota Corolla    1  1    4
Toyota Corona     1  0    3
Dodge Challenger  0  0    3
AMC Javelin       0  0    3
Camaro Z28        0  0    3
Pontiac Firebird  0  0    3
Fiat X1-9         1  1    4
Porsche 914-2     0  1    5
Lotus Europa      1  1    5
Ford Pantera L    0  1    5
Ferrari Dino      0  1    5
```

```
# select row 5 only
mtcars[5,]
```

```
                   mpg cyl disp  hp drat   wt  qsec vs am gear carb
Hornet Sportabout 18.7   8  360 175 3.15 3.44 17.02  0  0    3    2
                         car_brand
Hornet Sportabout Hornet Sportabout
```

```
# select column 2 and 6 only
mtcars[c(2,6)]
```

```
                    cyl    wt
Mazda RX4             6 2.620
Mazda RX4 Wag         6 2.875
Datsun 710            4 2.320
Hornet 4 Drive        6 3.215
Hornet Sportabout     8 3.440
Valiant               6 3.460
Duster 360            8 3.570
Merc 240D             4 3.190
Merc 230              4 3.150
Merc 280              6 3.440
Merc 280C             6 3.440
Merc 450SE            8 4.070
Merc 450SL            8 3.730
Merc 450SLC           8 3.780
Cadillac Fleetwood    8 5.250
Lincoln Continental   8 5.424
Chrysler Imperial     8 5.345
Fiat 128              4 2.200
Honda Civic           4 1.615
Toyota Corolla        4 1.835
Toyota Corona         4 2.465
Dodge Challenger      8 3.520
AMC Javelin           8 3.435
Camaro Z28            8 3.840
Pontiac Firebird      8 3.845
Fiat X1-9             4 1.935
Porsche 914-2         4 2.140
Lotus Europa          4 1.513
Ford Pantera L        8 3.170
Ferrari Dino          6 2.770
Maserati Bora         8 3.570
Volvo 142E            4 2.780
```

```
mtcars[,c(2,6)]
```

```
                   cyl    wt
Mazda RX4            6 2.620
Mazda RX4 Wag        6 2.875
Datsun 710           4 2.320
Hornet 4 Drive       6 3.215
Hornet Sportabout    8 3.440
Valiant              6 3.460
Duster 360           8 3.570
Merc 240D            4 3.190
Merc 230             4 3.150
Merc 280             6 3.440
Merc 280C            6 3.440
Merc 450SE           8 4.070
Merc 450SL           8 3.730
Merc 450SLC          8 3.780
Cadillac Fleetwood   8 5.250
Lincoln Continental  8 5.424
Chrysler Imperial    8 5.345
Fiat 128             4 2.200
Honda Civic          4 1.615
Toyota Corolla       4 1.835
Toyota Corona        4 2.465
Dodge Challenger     8 3.520
AMC Javelin          8 3.435
Camaro Z28           8 3.840
Pontiac Firebird     8 3.845
Fiat X1-9            4 1.935
Porsche 914-2        4 2.140
Lotus Europa         4 1.513
Ford Pantera L       8 3.170
Ferrari Dino         6 2.770
Maserati Bora        8 3.570
Volvo 142E           4 2.780
```

```
mtcars[c(2,6),]
```

```
              mpg cyl disp  hp drat    wt  qsec vs am gear carb     car_brand
Mazda RX4 Wag 21.0   6  160 110 3.90 2.875 17.02  0  1    4    4 Mazda RX4 Wag
Valiant       18.1   6  225 105 2.76 3.460 20.22  1  0    3    1       Valiant
```

```
# select columns 2 to 6
mtcars[2:6]
```

```
                   cyl  disp  hp drat    wt
Mazda RX4            6 160.0 110 3.90 2.620
Mazda RX4 Wag        6 160.0 110 3.90 2.875
Datsun 710           4 108.0  93 3.85 2.320
Hornet 4 Drive       6 258.0 110 3.08 3.215
Hornet Sportabout    8 360.0 175 3.15 3.440
Valiant              6 225.0 105 2.76 3.460
Duster 360           8 360.0 245 3.21 3.570
Merc 240D            4 146.7  62 3.69 3.190
Merc 230             4 140.8  95 3.92 3.150
Merc 280             6 167.6 123 3.92 3.440
Merc 280C            6 167.6 123 3.92 3.440
Merc 450SE           8 275.8 180 3.07 4.070
Merc 450SL           8 275.8 180 3.07 3.730
Merc 450SLC          8 275.8 180 3.07 3.780
Cadillac Fleetwood   8 472.0 205 2.93 5.250
Lincoln Continental  8 460.0 215 3.00 5.424
Chrysler Imperial    8 440.0 230 3.23 5.345
Fiat 128             4  78.7  66 4.08 2.200
Honda Civic          4  75.7  52 4.93 1.615
Toyota Corolla       4  71.1  65 4.22 1.835
```

```
Toyota Corona         4 120.1  97 3.70 2.465
Dodge Challenger      8 318.0 150 2.76 3.520
AMC Javelin           8 304.0 150 3.15 3.435
Camaro Z28            8 350.0 245 3.73 3.840
Pontiac Firebird      8 400.0 175 3.08 3.845
Fiat X1-9             4  79.0  66 4.08 1.935
Porsche 914-2         4 120.3  91 4.43 2.140
Lotus Europa          4  95.1 113 3.77 1.513
Ford Pantera L        8 351.0 264 4.22 3.170
Ferrari Dino          6 145.0 175 3.62 2.770
Maserati Bora         8 301.0 335 3.54 3.570
Volvo 142E            4 121.0 109 4.11 2.780
```

```r
# select all the cars that have 4 cylinders
m <- mtcars[mtcars$cyl == 4,]
length(m)
```

```
[1] 12
```

```r
length(mtcars[,1])
```

```
[1] 32
```

```r
nrow(mtcars)
```

```
[1] 32
```

```r
ncol(mtcars)
```

```
[1] 12
```

```r
# how many cars have automatic transmission
nrow(mtcars[mtcars$am == 0,])
```

```
[1] 19
```

```r
# select "Fiat 128" row by its rowname
# mtcars["Fiat 128"]
mtcars["Fiat 128",]
```

```
          mpg cyl disp hp drat  wt  qsec vs am gear carb car_brand
Fiat 128 32.4   4 78.7 66 4.08 2.2 19.47  1  1    4    1  Fiat 128
```

```r
mean(c(40,60,40,60))
```

```
[1] 50
```

```r
mtcars[mtcars$mpg < mean(mtcars$mpg),]
```

```
                   mpg cyl  disp  hp drat    wt  qsec vs am gear carb
Hornet Sportabout  18.7   8 360.0 175 3.15 3.440 17.02  0  0    3    2
Valiant            18.1   6 225.0 105 2.76 3.460 20.22  1  0    3    1
Duster 360         14.3   8 360.0 245 3.21 3.570 15.84  0  0    3    4
Merc 280           19.2   6 167.6 123 3.92 3.440 18.30  1  0    4    4
Merc 280C          17.8   6 167.6 123 3.92 3.440 18.90  1  0    4    4
Merc 450SE         16.4   8 275.8 180 3.07 4.070 17.40  0  0    3    3
Merc 450SL         17.3   8 275.8 180 3.07 3.730 17.60  0  0    3    3
Merc 450SLC        15.2   8 275.8 180 3.07 3.780 18.00  0  0    3    3
Cadillac Fleetwood 10.4   8 472.0 205 2.93 5.250 17.98  0  0    3    4
Lincoln Continental 10.4  8 460.0 215 3.00 5.424 17.82  0  0    3    4
Chrysler Imperial  14.7   8 440.0 230 3.23 5.345 17.42  0  0    3    4
Dodge Challenger   15.5   8 318.0 150 2.76 3.520 16.87  0  0    3    2
AMC Javelin        15.2   8 304.0 150 3.15 3.435 17.30  0  0    3    2
Camaro Z28         13.3   8 350.0 245 3.73 3.840 15.41  0  0    3    4
```

```
Pontiac Firebird    19.2   8 400.0 175 3.08 3.845 17.05  0  0    3    2
Ford Pantera L      15.8   8 351.0 264 4.22 3.170 14.50  0  1    5    4
Ferrari Dino        19.7   6 145.0 175 3.62 2.770 15.50  0  1    5    6
Maserati Bora       15.0   8 301.0 335 3.54 3.570 14.60  0  1    5    8
                           car_brand
Hornet Sportabout     Hornet Sportabout
Valiant                         Valiant
Duster 360                   Duster 360
Merc 280                       Merc 280
Merc 280C                     Merc 280C
Merc 450SE                   Merc 450SE
Merc 450SL                   Merc 450SL
Merc 450SLC                 Merc 450SLC
Cadillac Fleetwood   Cadillac Fleetwood
Lincoln Continental Lincoln Continental
Chrysler Imperial     Chrysler Imperial
Dodge Challenger       Dodge Challenger
AMC Javelin                 AMC Javelin
Camaro Z28                   Camaro Z28
Pontiac Firebird       Pontiac Firebird
Ford Pantera L           Ford Pantera L
Ferrari Dino               Ferrari Dino
Maserati Bora             Maserati Bora
```

Select a single row

```r
# print(): displays the supplied value or object.
# A data frame uses [row, column] indexing to select records and variables.
# A blank row or column position keeps all entries in that dimension.
first_row <- mtcars[1, ]
print(first_row)
```

```
          mpg cyl disp  hp drat   wt  qsec vs am gear carb car_brand
Mazda RX4  21   6  160 110  3.9 2.62 16.46  0  1    4    4 Mazda RX4
```

Select a specific element

```r
# print(): displays the supplied value or object.
# A data frame uses [row, column] indexing to select records and variables.
# A blank row or column position keeps all entries in that dimension.
# Accessing the Age of the second person
mpg <- mtcars[2, "mpg"]
print(mpg)
```

```
[1] 21
```

Using `subset()` Function

The **`subset()`** function can be used for more complex subsetting based on conditions.

- **Select rows where Age is greater than 25**:

  ```r
  # print(): displays the supplied value or object.
  # subset() selects observations satisfying the supplied condition.
  # Conditions refer to column names within the selected data frame; check missing
   ↪  values separately.
  more_than_4_cyl <- subset(mtcars, cyl > 4)
  print(more_than_4_cyl)
  ```

  ```
                      mpg cyl  disp  hp drat    wt  qsec vs am gear carb
  Mazda RX4           21.0   6 160.0 110 3.90 2.620 16.46  0  1    4    4
  Mazda RX4 Wag       21.0   6 160.0 110 3.90 2.875 17.02  0  1    4    4
  Hornet 4 Drive      21.4   6 258.0 110 3.08 3.215 19.44  1  0    3    1
  Hornet Sportabout   18.7   8 360.0 175 3.15 3.440 17.02  0  0    3    2
  Valiant             18.1   6 225.0 105 2.76 3.460 20.22  1  0    3    1
  Duster 360          14.3   8 360.0 245 3.21 3.570 15.84  0  0    3    4
  Merc 280            19.2   6 167.6 123 3.92 3.440 18.30  1  0    4    4
  Merc 280C           17.8   6 167.6 123 3.92 3.440 18.90  1  0    4    4
  ```

```
     Merc 450SE          16.4   8 275.8 180 3.07 4.070 17.40  0  0    3    3
     Merc 450SL          17.3   8 275.8 180 3.07 3.730 17.60  0  0    3    3
     Merc 450SLC         15.2   8 275.8 180 3.07 3.780 18.00  0  0    3    3
     Cadillac Fleetwood  10.4   8 472.0 205 2.93 5.250 17.98  0  0    3    4
     Lincoln Continental 10.4   8 460.0 215 3.00 5.424 17.82  0  0    3    4
     Chrysler Imperial   14.7   8 440.0 230 3.23 5.345 17.42  0  0    3    4
     Dodge Challenger    15.5   8 318.0 150 2.76 3.520 16.87  0  0    3    2
     AMC Javelin         15.2   8 304.0 150 3.15 3.435 17.30  0  0    3    2
     Camaro Z28          13.3   8 350.0 245 3.73 3.840 15.41  0  0    3    4
     Pontiac Firebird    19.2   8 400.0 175 3.08 3.845 17.05  0  0    3    2
     Ford Pantera L      15.8   8 351.0 264 4.22 3.170 14.50  0  1    5    4
     Ferrari Dino        19.7   6 145.0 175 3.62 2.770 15.50  0  1    5    6
     Maserati Bora       15.0   8 301.0 335 3.54 3.570 14.60  0  1    5    8
                                      car_brand
     Mazda RX4                        Mazda RX4
     Mazda RX4 Wag                Mazda RX4 Wag
     Hornet 4 Drive              Hornet 4 Drive
     Hornet Sportabout       Hornet Sportabout
     Valiant                           Valiant
     Duster 360                    Duster 360
     Merc 280                        Merc 280
     Merc 280C                      Merc 280C
     Merc 450SE                    Merc 450SE
     Merc 450SL                    Merc 450SL
     Merc 450SLC                  Merc 450SLC
     Cadillac Fleetwood     Cadillac Fleetwood
     Lincoln Continental Lincoln Continental
     Chrysler Imperial       Chrysler Imperial
     Dodge Challenger         Dodge Challenger
     AMC Javelin                   AMC Javelin
     Camaro Z28                     Camaro Z28
     Pontiac Firebird         Pontiac Firebird
     Ford Pantera L             Ford Pantera L
     Ferrari Dino                 Ferrari Dino
     Maserati Bora               Maserati Bora
```

Filtering data frames

```r
# c(): combines values into a vector, converting them to a common type when needed.
# print(): displays the supplied value or object.
# data.frame() combines equally aligned vectors into a rectangular table.
# Columns describe variables and rows describe observations; retain the meaning and units
 ↪  of each column.
# Example of creating a data.frame
name <- c("John", "Alice", "Bob", "Eva")
age <- c(23, 25, 30, 22)
gender <- c("Male", "Female", "Male", "Female")

# Combine vectors into a data.frame
students <- data.frame(name, age, gender)

# Display the data.frame
print(students)
```

```
   name age gender
1  John  23   Male
2 Alice  25 Female
3   Bob  30   Male
4   Eva  22 Female
```

```r
# Accessing a column using $
print(students$age)
```

```
[1] 23 25 30 22
```

```r
# Accessing a column using indexing
print(students["gender"])
```

```
  gender
1   Male
2 Female
3   Male
4 Female
```

```r
# Access the element in the 2nd row, 1st column
print(students[2, 1])
```

```
[1] "Alice"
```

```r
# Access the entire 3rd row
print(students[3, ])
```

```
  name age gender
3  Bob  30   Male
```

```r
# Access the entire 2nd column
print(students[, 2])
```

```
[1] 23 25 30 22
```

```r
# Filter rows where age is greater than 23
older_students <- students[students$age > 23, ]

# Display the filtered data.frame
print(older_students)
```

```
   name age gender
2 Alice  25 Female
3   Bob  30   Male
```

Miscellaneous Related Topics - عيضاوم ةقرفتم ذات ةقلع

Printing Output

In R, you have several functions at your disposal to print output. Here are the most commonly used ones:

1. **`print()`** - This is the basic R function to print objects to the console. It's a generic function, which means that it can have different methods for different object types.

   ```r
   # print() displays the supplied value or the contents of a saved object.
   # Use <- to assign a result to a name before reusing it in later expressions.
   print("Hello, World!")
   ```

   ```
   [1] "Hello, World!"
   ```

2. **`cat()`** - Concatenates and prints objects. It is often used for printing character strings and does not automatically add a newline unless you specify it with `"\n"`.

   ```r
   # cat() writes values together as console output without the usual vector labels.
   # Use "\n" to insert a new line; this displays text rather than returning a joined
    ↪  character vector.
   cat("Hello,", "World!", "\n")
   ```

   ```
   Hello, World!
   ```

3. The **paste** function in R is used to concatenate strings. Unlike `print` or `cat`, `paste` does not print to the console but returns a single string that is the concatenation of its arguments. The `paste` function is particularly useful when you want to combine strings and variables to create a new string.

Here's a basic usage of `paste`:

```
# paste() joins values into character strings.
# sep controls the separator between inputs; collapse joins elements of a vector into one
 ↪  string.
# This will not print to the console but will return a string.
result <- paste("Hello", "World", sep = " ")

# To print the result, you can use print() or cat()
print(result)  # This will print "Hello World" to the console
```

```
[1] "Hello World"
```

```
cat(result, "\n")  # This will also print "Hello World", followed by a new line
```

```
Hello World
```

Requesting user input

Test this command in the console window not in quarto.

```
# readline(): reads one line of interactive console input as text; it is not suitable for
 ↪  unattended input.
# cat() writes values together as console output without the usual vector labels.
# Use "\n" to insert a new line; this displays text rather than returning a joined
 ↪  character vector.
name <- readline("Please enter your name: ")
cat("Your name is:", name)
```

Values Assignments in R

We can use different syntax to make an assignment in R.

The preferred syntax is to use the Values Gets e.g. x <- 10, which means x Gets the value 10.

We can also use the conventional assignment operator.

The Gets operator has the advantage of escaping the ambiguity of the assignment and equality operators = vs ==

```
# Assignments store a value in an object named on the receiving side of <-, ->, or =.
# The final 1 = x line is intentionally invalid because a literal number cannot receive an
 ↪  assignment.
x <- 1 # valid, x gets 1
x
```

```
[1] 1
```

```
1 -> x # valid, x gets 1
x
```

```
[1] 1
```

```r
x = 1 # valid, x = 1
x
```

```
[1] 1
```

```r
1 = x # invalid, x
```

```
Error in `1 = x`:
! invalid (do_set) left-hand side to assignment
```

List Memory, Clean Memory

```r
# rm() removes named objects from the current R environment.
# ls() lists object names; rm(list=ls()) clears them, so use that form only deliberately.
# list variables
x <- 1
y <- 2
ls()
```

```
 [1] "a"                 "age"               "ages"
 [4] "answers"           "array_data"        "b"
 [7] "banana_count"      "BMI"               "c"
[10] "car_names"         "colors"            "combined_seq"
[13] "complex_num"       "count"             "d"
[16] "datsun_710"        "de_vectorized_time" "df"
[19] "exclude_range"     "exclude_second"    "filtered_numbers"
[22] "first_car_name"    "first_row"         "fruits"
[25] "gender"            "i"                 "is_valid"
[28] "list_data"         "m"                 "matrix_data"
[31] "miles_per_hour"    "more_than_4_cyl"   "mpg"
[34] "mtcars"            "myString"          "N"
[37] "name"              "numbers"           "older_students"
[40] "pandoc_dir"        "quarto_bin_path"   "range_numbers"
[43] "rep_each"          "rep_times"         "result"
[46] "selected_numbers"  "sentence"          "seq_1_to_10"
[49] "seq_by_2"          "seq_length_out"    "some_fruits"
[52] "students"          "subset_df"         "test_result"
[55] "third_number"      "vectorized_time"   "weight"
[58] "x"                 "y"
```

```r
# clear specific values
rm(x,y)
ls()
```

```
 [1] "a"                 "age"               "ages"
 [4] "answers"           "array_data"        "b"
 [7] "banana_count"      "BMI"               "c"
[10] "car_names"         "colors"            "combined_seq"
[13] "complex_num"       "count"             "d"
[16] "datsun_710"        "de_vectorized_time" "df"
[19] "exclude_range"     "exclude_second"    "filtered_numbers"
[22] "first_car_name"    "first_row"         "fruits"
[25] "gender"            "i"                 "is_valid"
[28] "list_data"         "m"                 "matrix_data"
[31] "miles_per_hour"    "more_than_4_cyl"   "mpg"
[34] "mtcars"            "myString"          "N"
[37] "name"              "numbers"           "older_students"
[40] "pandoc_dir"        "quarto_bin_path"   "range_numbers"
[43] "rep_each"          "rep_times"         "result"
[46] "selected_numbers"  "sentence"          "seq_1_to_10"
[49] "seq_by_2"          "seq_length_out"    "some_fruits"
[52] "students"          "subset_df"         "test_result"
[55] "third_number"      "vectorized_time"   "weight"
```

```
# clear all values
rm(list = ls())
ls()
```

```
character(0)
```

- We can also click on the “Broom” icon in the Environment tab to clear all objects.
- Alternatively, use the **`Session > Clear Workspace...`** menu option.
- Finally, restarting the R session is a surefire way to clear all objects and start fresh. In RStudio, you can do this via the menu **`Session > Restart R`** or by using the **`Ctrl+Shift+F10`** shortcut on Windows/Linux or **`Cmd+Shift+F10`** on macOS.

Flow of Control

Repetition in R (Loops)

Loops are fundamental constructs in programming that allow you to execute a block of code repeatedly. In R, the primary looping constructs are the `for` loop, the `while` loop, and the `repeat` loop. This section will guide you through the syntax and usage of each, with examples to illustrate how they can be used in data analysis tasks.

The `for` Loop

The `for` loop is used to iterate over a sequence or a vector and execute a block of code for each element.

Syntax:

```
# A for loop repeats its body once for each value in the supplied sequence.
# The loop variable identifies the current value or position; vectorized arithmetic can often replace the loop.
for (variable in sequence) {
  # Code to execute for each element of the sequence
}
```

Example:

Let’s iterate over a vector of numbers and print each number squared.

```
# print(): displays the supplied value or object.
# A for loop repeats its body once for each value in the supplied sequence.
# The loop variable identifies the current value or position; vectorized arithmetic can often replace the loop.
# Create a vector of numbers
numbers <- 1:5

# Use a for loop to iterate and print squares
for (num in numbers) {
  print(num^2)
}
```

```
[1] 1
[1] 4
[1] 9
[1] 16
[1] 25
```

What are the correct values for the *s in the code below so that it will iterate over all the items of x array?

```
# This unfinished loop is a practice template: replace *:* with suitable integer
 ↪  positions.
# Use x[i] to select each position and keep the exercise unevaluated until the loop is
 ↪  complete.
x <- 50:60
for (i in *:*)
{
  print(x[i])
}
```

In this example, the loop runs five times, squaring each number from 1 to 5.

The `while` Loop

The `while` loop continues to execute as long as a specified condition is true. It's useful when the number of iterations is not known beforehand.

Syntax:

```
# while() repeats its body as long as its condition remains TRUE.
# Update the value used in the condition so a running example can eventually stop.
while (condition) {
  # Code to execute as long as the condition is true
}
```

Example:

We will use a `while` loop to simulate a countdown.

```
# print(): displays the supplied value or object.
# paste(): joins text using sep between pieces; collapse combines a vector into one
 ↪  string.
# while() repeats its body as long as its condition remains TRUE.
# Update the value used in the condition so a running example can eventually stop.
# Set the starting point of the countdown
countdown <- 5

# Begin the while loop
while (countdown > 0) {
  print(paste("Counting down:", countdown))
  countdown <- countdown - 1
}
```

```
[1] "Counting down: 5"
[1] "Counting down: 4"
[1] "Counting down: 3"
[1] "Counting down: 2"
[1] "Counting down: 1"
```

```
# When the loop ends
print("Lift off!")
```

```
[1] "Lift off!"
```

This loop will run until `countdown` is no longer greater than 0.

The repeat Loop

The repeat loop executes an indefinite number of times until explicitly told to stop with a break statement. It's the most basic form of loop and is less commonly used than for and while loops due to its potential to create infinite loops if not handled carefully.

Syntax:

```
# repeat executes its body until an explicit break exits the loop.
# The if condition decides when to stop; update the working value on each unsuccessful
 ↪  pass.
repeat {
  # Code to execute indefinitely
  if (stop_condition) {
    break
  }
}
```

Example:

Here's a repeat loop that finds the first number divisible by 13 greater than 100.

```
# print(): displays the supplied value or object.
# paste(): joins text using sep between pieces; collapse combines a vector into one
 ↪  string.
# repeat executes its body until an explicit break exits the loop.
# The if condition decides when to stop; update the working value on each unsuccessful
 ↪  pass.
# Initialize the number
number <- 101

# Start the repeat loop
repeat {
  # Check if the number is divisible by 13
  if (number %% 13 == 0) {
    print(paste("Found a number divisible by 13:", number))
    break
  }
  # Increment the number
  number <- number + 1
}
```

```
[1] "Found a number divisible by 13: 104"
```

The loop runs indefinitely, incrementing number until it finds one that satisfies the condition.

Nested Loops

Loops can be nested inside other loops, which is useful for iterating over multiple dimensions, such as rows and columns in a matrix.

Example:

```
# matrix(): arranges values in rows and columns; byrow=TRUE fills rows rather than the
 ↪  default columns.
# nrow(): returns the number of rows in a table or matrix.
# ncol(): returns the number of columns in a table or matrix.
# print(): displays the supplied value or object.
# A for loop repeats its body once for each value in the supplied sequence.
```

```r
# The loop variable identifies the current value or position; vectorized arithmetic can
 ↪  often replace the loop.
# Create a 3x3 matrix
my_matrix <- matrix(1:9, nrow = 3)

# Iterate over each element of the matrix
for (row in 1:nrow(my_matrix)) {
  for (col in 1:ncol(my_matrix)) {
    # Multiply each element by 2
    my_matrix[row, col] <- my_matrix[row, col] * 2
  }
}
print(my_matrix)
```

```
     [,1] [,2] [,3]
[1,]    2    8   14
[2,]    4   10   16
[3,]    6   12   18
```

In this nested loop example, every element of `my_matrix` is doubled.

Conditional Branching in R

Introduction to Conditional Branching

Conditional branching in R allows you to control the flow of execution based on conditions. It's a fundamental concept in programming that lets you execute certain sections of code while skipping others, depending on whether specified conditions are TRUE or FALSE. This section will explain how to use conditional statements in R, such as `if`, `else`, and `else if`, along with practical examples.

The `if` Statement

The `if` statement is the simplest form of conditional branching. It evaluates a condition and, if that condition is TRUE, it executes a block of code.

Syntax:

```r
# if evaluates a condition and runs its body when the condition is TRUE.
# else or else if handles alternative cases; replace placeholder conditions before
 ↪  running a template.
if (condition) {
  # Code to execute if the condition is TRUE
}
```

Example:

Here, we check if a number is positive.

```r
# print(): displays the supplied value or object.
# if evaluates a condition and runs its body when the condition is TRUE.
# else or else if handles alternative cases; replace placeholder conditions before
 ↪  running a template.
number <- 5

if (number > 0) {
  print("The number is positive.")
}
```

```
[1] "The number is positive."
```

The `else` Statement

The `else` statement is used together with an `if` statement to execute code when the `if` condition is not TRUE.

Syntax:

```
# if evaluates a condition and runs its body when the condition is TRUE.
# else or else if handles alternative cases; replace placeholder conditions before
 ↪  running a template.
if (condition) {
  # Code to execute if the condition is TRUE
} else {
  # Code to execute if the condition is FALSE
}
```

Example:

Expanding on the previous example, we also handle the case when the number is not positive.

```
# print(): displays the supplied value or object.
# if evaluates a condition and runs its body when the condition is TRUE.
# else or else if handles alternative cases; replace placeholder conditions before
 ↪  running a template.
number <- -3

if (number > 0) {
  print("The number is positive.")
} else {
  print("The number is not positive.")
}
```

```
[1] "The number is not positive."
```

The `else if` Statement

The `else if` statement allows you to check multiple conditions. If the condition for `if` is FALSE, it checks the condition for `else if`, and so on.

Syntax:

```
# if evaluates a condition and runs its body when the condition is TRUE.
# else or else if handles alternative cases; replace placeholder conditions before
 ↪  running a template.
if (condition1) {
  # Code to execute if condition1 is TRUE
} else if (condition2) {
  # Code to execute if condition1 is FALSE and condition2 is TRUE
} else {
  # Code to execute if none of the above conditions are TRUE
}
```

Example:

Here, we classify a number as positive, negative, or zero.

```
# print(): displays the supplied value or object.
# if evaluates a condition and runs its body when the condition is TRUE.
# else or else if handles alternative cases; replace placeholder conditions before
 ↪  running a template.
number <- 0

if (number > 0) {
  print("The number is positive.")
} else if (number < 0) {
  print("The number is negative.")
} else {
  print("The number is zero.")
}
```

```
[1] "The number is zero."
```

The `switch` Statement

The `switch` statement is a multi-way branch used when you want to compare the same variable (or expression) with many different possible values and execute different pieces of code for each value.

Syntax:

```
# switch() chooses a branch using the supplied expression or character label.
# The final unnamed branch can provide a fallback when none of the named labels matches.
switch(EXPR,
       "case1" = {# Code for case 1},
       "case2" = {# Code for case 2},
       # ...,
       {# Default case code}
)
```

Example:

The following example shows how `switch` can be used to print different messages based on the value of a character variable.

```
# print(): displays the supplied value or object.
# switch() chooses a branch using the supplied expression or character label.
# The final unnamed branch can provide a fallback when none of the named labels matches.
day <- "Tue"

switch(day,
       "Mon" = {print("Start of the work week")},
       "Tue" = {print("Second day of the work week")},
       "Wed" = {print("Middle of the work week")},
       "Thu" = {print("Approaching the end")},
       "Fri" = {print("Last day of the work week")},
       {print("It must be the weekend")}
)
```

```
[1] "Second day of the work week"
```

Creating Functions in R

Introduction

Functions in R are crucial for efficient and organized code, allowing for repetitive tasks to be encapsulated into single, callable entities. This chapter demonstrates how to craft your own functions, enabling the automation of tasks and making your code more modular and readable.

Basic Function Structure

A function is defined using the `function` keyword, followed by parentheses containing any arguments, and a body enclosed in curly braces.

Example: Simple Addition Function

```
# print(): displays the supplied value or object.
# add_two_numbers(): calls the function defined in this lesson to return the sum of its
 ↪  two inputs.
# function() defines a reusable calculation with named input arguments.
# The body computes a result for those inputs; return() explicitly chooses the value
 ↪  passed back.
add_two_numbers <- function(a, b) {
  result <- a + b
  return(result)
}

# Using the function
print(add_two_numbers(5, 3))
```

```
[1] 8
```

Function Arguments and Return Values

Functions can take arguments and return results. R functions return the result of the last line in the function body by default, but the `return()` function can be used for clarity or early returns.

Example: Greeting Function

```
# paste(): joins text using sep between pieces; collapse combines a vector into one
 ↪  string.
# print(): displays the supplied value or object.
# greet_person(): calls the locally defined function to format a greeting from a name.
# function() defines a reusable calculation with named input arguments.
# The body computes a result for those inputs; return() explicitly chooses the value
 ↪  passed back.
greet_person <- function(name) {
  greeting <- paste("Hello,", name)
  return(greeting)
}

# Using the function
print(greet_person("Alice"))
```

```
[1] "Hello, Alice"
```

Default Argument Values

You can specify default values for arguments, making them optional.

Example: Power Function

```r
# print(): displays the supplied value or object.
# raise_to_power(): calls the locally defined function using its supplied or default
 ↪  exponent.
# function() defines a reusable calculation with named input arguments.
# The body computes a result for those inputs; return() explicitly chooses the value
 ↪  passed back.
raise_to_power <- function(base, exponent = 2) {
  result <- base ^ exponent
  return(result)
}

# Using the function with default exponent
print(raise_to_power(4))
```

```
[1] 16
```

```r
# Specifying both arguments
print(raise_to_power(4, 3))
```

```
[1] 64
```

Returning Multiple Values

Functions in R can return multiple values using a list.

Example: Statistics Function

```r
# mean(): computes an average; logical TRUE/FALSE values contribute 1/0, giving a
 ↪  proportion.
# sum(): adds vector entries; on logical values it counts TRUE entries.
# list(): stores named or unnamed objects that may have different types and lengths.
# calculate_stats(): calls the locally defined function to return the summaries listed in
 ↪  its body.
# c(): combines values into a vector, converting them to a common type when needed.
# print(): displays the supplied value or object.
# function() defines a reusable calculation with named input arguments.
# The body computes a result for those inputs; return() explicitly chooses the value
 ↪  passed back.
calculate_stats <- function(numbers) {
  mean_val <- mean(numbers)
  sum_val <- sum(numbers)
  return(list(mean = mean_val, sum = sum_val))
}

# Using the function
stats <- calculate_stats(c(1, 2, 3, 4, 5))
print(stats)
```

```
$mean
[1] 3

$sum
[1] 15
```

Anonymous Functions

For short, one-time operations, anonymous functions can be used.

Example: Doubling Numbers

```
# print(): displays the supplied value or object.
# sapply() applies a calculation to each input element and simplifies compatible results
 ↪  to a vector.
# function(x) x * 2 defines the operation: double each element without changing the input
 ↪  vector.
numbers <- 1:5
doubled_numbers <- sapply(numbers, function(x) x * 2)
print(doubled_numbers)
```

```
[1]  2  4  6  8 10
```

Scope and Environment

Variables defined inside a function are local to that function.

Example: Increment Function

```
# increment(): calls the locally defined function to increase its input as specified in
 ↪  its body.
# print(): displays the supplied value or object.
# function() defines a reusable calculation with named input arguments.
# The body computes a result for those inputs; return() explicitly chooses the value
 ↪  passed back.
increment <- function(x) {
  y <- x + 1
  return(y)
}

incremented_value <- increment(5)
print(incremented_value)
```

```
[1] 6
```

```
# Trying to print 'y' here would result in an error because 'y' is not defined in this
 ↪  scope.
```